Customer Behavior Analytics Pipeline

A robust architectural flow for categorizing financial transactions, aggregating attributes, computing dynamic behavior metrics, and building predictive ML models.

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Phase 1: Data Ingestion

The primary high-volume data streams containing raw records of customer activities.

Financial Transactions

Raw records of customer spending, transfers, and point-of-sale activities.

- Raw text descriptions
- Transaction amounts & timestamps
- Merchant identification codes

Deposit Records

Periodic snapshots detailing account funding and overall liquidity events.

- Inbound fund origins
- Funding frequency (recurring vs. ad-hoc)
- Deposit magnitude & variation

Phase 2: Categorization

Raw transaction descriptions are processed through a rules-based engine placing each transaction into a definitive category.

Pattern Recognition

Identifies whether or not a transaction is recurring based on established historical algorithms.

Standardization

Maps transactions to definitive market categories (e.g., Groceries, Utilities, Rent).

Anomaly Detection

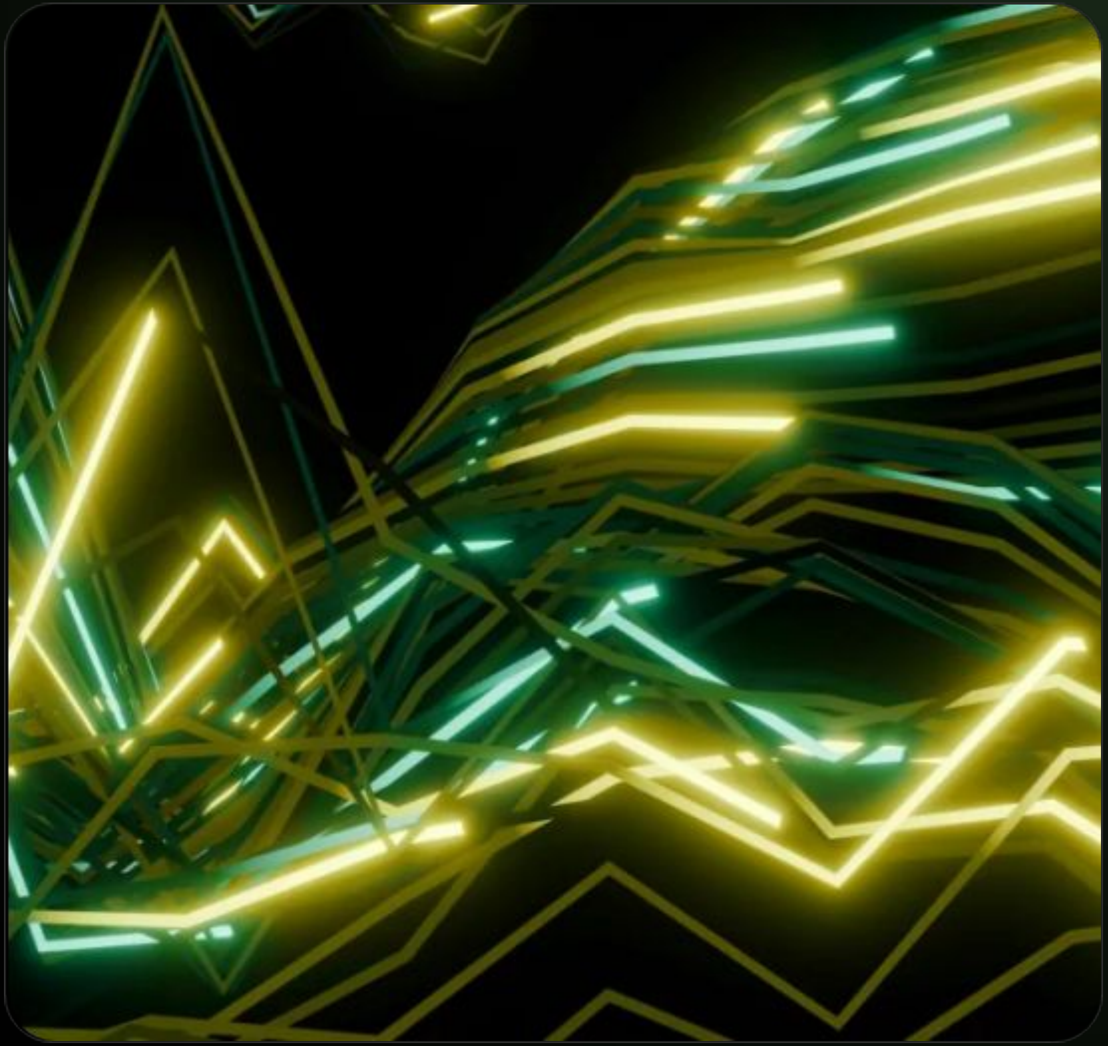
Flags unusual spending patterns that deviate from established customer history.

Data Enrichment

Prepares cleanly tagged and contextualized data for the subsequent aggregation layer.

Phase 3: Aggregation & Joining

- **Transaction Rollups** Categorized spending is grouped by month to compute average velocity and volume.
- **Attribute Aggregation** Attributes are aggregated along with deposit balances to calculate advanced behavioral metrics.
- **Feature Joining** Metrics and deposit summaries are unified via customer identifier to create a single master record.



Phase 4:
Metrics Output

Time Series

Correlates spending patterns with deposit liquidity to create a continuous behavior time series for predictive modeling.

Phase 5: AutoML Modeling

After creating the behavior time series, Automated Machine Learning was utilized to rapidly test different approaches for modeling customer risk and business opportunity.

XGBoost

Gradient boosted decision trees optimized for high-performance classification of behavioral risk factors.

GLM

Generalized Linear Models providing high interpretability required for regulatory compliance in risk scoring.

LightGBM

Highly efficient gradient boosting framework handling large-scale time series data with fast training speeds.

Ensemble

Stacked ensemble methods combining predictions from multiple algorithms to achieve superior accuracy.

End-to-End Pipeline Flowchart

From raw ingestion to predictive modeling

1. Extract
Ingest raw records

2. Categorize
Apply rule mapping

3. Aggregate
Monthly metric rollups

4. Time Series
Compute trends

5. AutoML
Model risk factors

Implementation & Impact

- **Powered by PySpark** Engineered using Apache Spark's Python API to enable distributed, high-performance data transformations across cluster nodes.
- **Massive Scale** Optimized algorithms allowed the pipeline to ingest, categorize, and aggregate billions of transactions in just a few hours.
- **Actionable Intelligence** The pipeline directly supplies critical behavioral features to dynamic risk rating and credit product opportunity models.

Image Sources

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